Business Storytelling with Data – Sales Performance Analysis and Recommendations

1. Introduction

Business storytelling with data is the process of examining business data, discovering important trends, and presenting insights that support decision-making.

In this task, a Sales Overview Dashboard created using Microsoft Power BI was analysed to understand sales performance across different cities, product categories, and months. The analysis focuses on revenue generation, transaction activity, estimated profit, and business improvement opportunities.

2. Objective

The main objectives of this task are:

- Analyse sales data through Power BI visualizations.
- Identify key insights from sales KPIs and charts.
- Compare sales performance across cities and product categories.
- Examine monthly revenue trends and estimated profitability.
- Develop practical business recommendations based on the findings.

3. Dataset Description

The sales dataset contains transaction records with the following attributes:

- Order ID
- Order Date
- Customer ID
- Product Category
- Product Name
- City
- Quantity
- Unit Price
- Sales Amount

An additional column called **Estimated Profit** was calculated using assumed profit margins for different product categories.

Note: Estimated profit values are calculated for learning purposes and may differ from actual business profits.

4. Dashboard Overview

The Sales Overview Dashboard was designed in Microsoft Power BI to present business performance through interactive visualizations.

The dashboard includes:

- Total Sales: **2.18M**
- Total Transactions: **92**
- Total Estimated Profit: **481.76K**
- Sum of Estimated Profit: **481.76K**
- City-wise Sales Analysis
- Total Sales by Category
- Monthly Sales Trend
- Total Estimated Profit by Category
- Interactive City and Date Filters

These visualizations provide a clear understanding of overall business performance.

5. Business Insights and Analysis

5.1 Insight 1 – City-wise Sales Performance

Observation:

Coimbatore generated the highest sales revenue of approximately 0.92M, followed by Madurai with 0.74M. Tiruchirappalli recorded comparatively lower sales of 0.52M.

Business Interpretation:

Coimbatore contributes the largest share of revenue among the three cities. The variation in sales indicates differences in regional demand and purchasing activity.

Business Recommendation:

Increase inventory availability and strengthen customer engagement in Coimbatore. Introduce targeted advertising and promotional campaigns in Tiruchirappalli to improve sales.

5.2 Insight 2 – Category-wise Sales Performance

Observation:

Furniture recorded the highest sales revenue of approximately 1.14M, followed by Electronics at 0.62M. Sports generated 0.26M, while Groceries contributed 0.16M.

Business Interpretation:

Furniture is the leading revenue-generating category, indicating stronger customer demand or higher transaction values. Groceries contribute the smallest share of total revenue.

Business Recommendation:

Maintain sufficient stock of furniture products and introduce attractive discounts or bundled offers for lower-performing categories.

5.3 Insight 3 – Monthly Sales Trend

Observation:

February recorded the highest monthly sales of approximately 0.46M, while April recorded the lowest at around 0.27M. Sales increased again during May and June.

Business Interpretation:

The monthly sales pattern indicates fluctuations in customer purchasing behaviour. The recovery after April suggests improving demand during the following months.

Business Recommendation:

Plan promotional activities during months with lower sales. Increase inventory and marketing efforts ahead of periods showing stronger customer demand.

5.4 Insight 4 – Estimated Profitability Analysis

Observation:

Furniture generated the highest estimated profit of approximately 0.38M, followed by Electronics at 0.21M. Sports contributed 0.12M, while Groceries recorded 0.07M.

Business Interpretation:

Furniture appears to be the strongest contributor to estimated profitability. Differences between categories may result from variations in sales volume and assumed profit margins.

Business Recommendation:

Focus on high-profit products while reviewing pricing strategies and operational expenses for categories with lower estimated profitability.

5.5 Insight 5 – Transaction and Revenue Analysis

Observation:

The dashboard shows 92 total transactions and approximately 2.18M in total sales.

The average revenue per transaction is:

Average Transaction Value = Total Sales / Total Transactions

= 2,180,000 / 92

= **23,695.65 (approximately)**

Business Interpretation:

The average transaction value helps measure the revenue generated from each transaction. This metric can be used to evaluate customer spending patterns and sales efficiency.

Business Recommendation:

Introduce product bundles, personalized discounts, and loyalty rewards to encourage repeat purchases and increase transaction value.

6. Interactive Dashboard Analysis

The dashboard includes City and Date slicers that allow users to explore sales performance under different filtering conditions.

By selecting individual cities and modifying date ranges, users can compare revenue, transaction counts, and estimated profitability.

These interactive features make it easier to identify regional differences and changes in business performance over time.

7. Key Findings

The analysis identified several important findings:

1. Coimbatore recorded the highest city-wise sales revenue.
2. Furniture was the leading category in both sales and estimated profit contribution.
3. February achieved the highest monthly sales, while April recorded the lowest.
4. Groceries contributed the least revenue among the displayed categories.
5. The dashboard recorded 92 transactions with an approximate average transaction value of 23.70K.

These findings provide useful information for improving sales planning and business strategies.

8. Business Recommendations

Based on the dashboard analysis, the following actions are recommended:

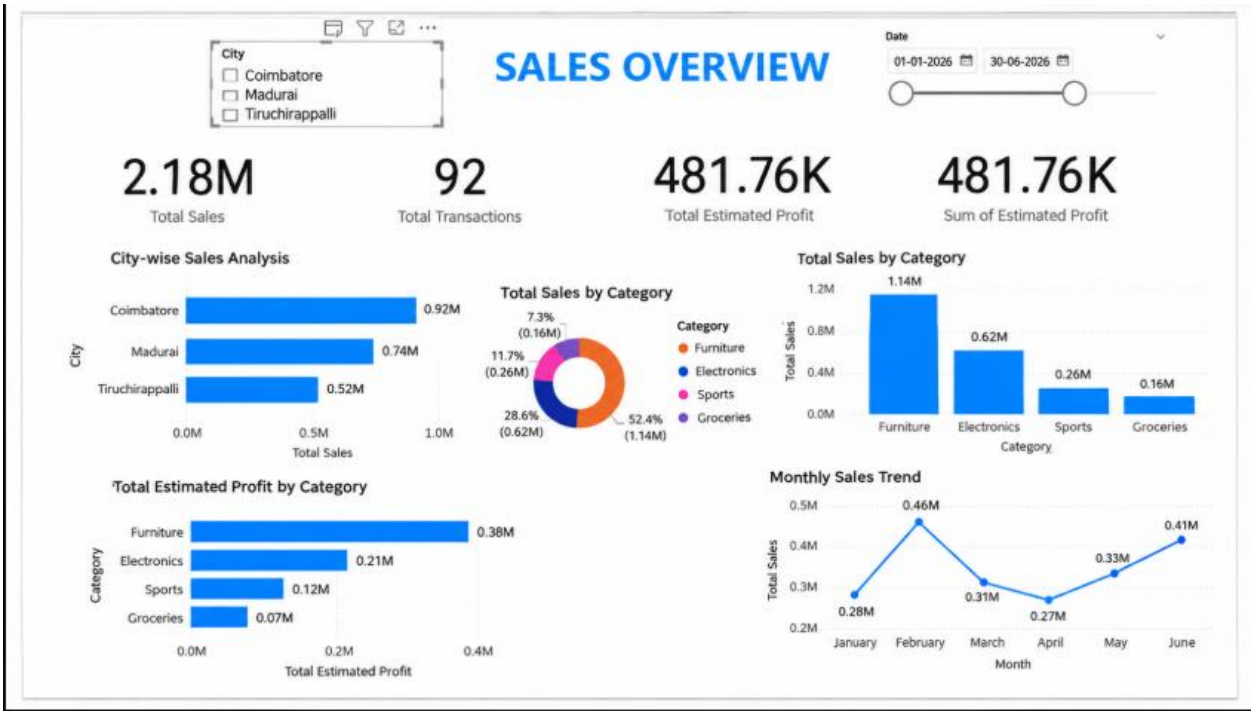
1. **Regional Marketing:** Improve promotional activities in cities with lower sales performance.
2. **Inventory Management:** Maintain adequate stock for high-demand categories, particularly Furniture.
3. **Seasonal Planning:** Schedule marketing campaigns based on monthly revenue fluctuations.
4. **Profit Optimization:** Monitor estimated profitability and evaluate pricing strategies across product categories.
5. **Customer Engagement:** Introduce loyalty programmes and bundled offers to encourage repeat purchases.

9. Conclusion

The Sales Overview Dashboard was successfully analysed using Microsoft Power BI to identify meaningful business insights.

The analysis highlighted differences in city-wise revenue, category performance, monthly sales patterns, transaction activity, and estimated profitability.

This exercise demonstrated how Power BI visualizations can transform sales data into understandable business insights and support informed decisions related to marketing, inventory management, and revenue growth.



CITY WISE ANALYSIS



Total Profit



Monthly Sales

